Technical Progress Report

Project name:	**Biped robot**		
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1. Executive Summary

This report summarizes the technical progress of the biped robot project this week. The core team has completed the setup and debugging of the visual recognition module (camera), achieving basic color recognition functionality, but the accuracy of shape recognition is insufficient, and black objects on a black desktop are prone to error recognition, which is a core optimization issue at this stage; after optimization, attempts were made to produce and assemble 3D printed structural components, but it was found that the motor could not be driven normally. Currently, fault 排查 work is underway and it is expected to be resolved next week; at the same time, the wiring of the motor / servo, pin definitions, and the initial development of an external virtual website to control the servo were completed, enabling basic remote control. In the future, the core will be the optimization of the camera recognition algorithm and the fault 排查 of the motor drive, while simultaneously advancing the mechanical assembly and iterative development of the virtual website functions.

2. Objectives & Scope

Complete the hardware setup and software debugging of the visual recognition module (camera) for the biped robot, achieving the recognition of the color and shape of the target object; complete the wiring and pin definitions of the robot's motors / steering gears, prepare the core hardware and complete the pre-test of the electrical system; complete the mechanical assembly of the optimized 3D printed structural components, test the basic drive and operation functions of the motors; develop an external virtual control website to realize the basic remote control of the robot's steering gears; lay the hardware and software foundation for subsequent motion control, system integration and algorithm optimization.

3. Technical Work Completed

3.1 Visual Recognition Module (Camera) Setup and Debugging

・Complete the hardware deployment of the ESP32-CAM camera module, achieve circuit connection with the main controller, power matching, and communication pin definition, and complete module initialization and normal startup;・Write and debug the image acquisition and preprocessing program, realize real-time image acquisition, transmission, and display, optimize camera exposure and brightness parameters, solve the problem of white screen and overexposure, and improve color reproduction;・Build an identification model based on the HSV color space, collect multi-color sample sets for training, achieve effective identification of pure colors and regular mixed-color objects, and control the response delay of identification;・Build a shape recognition framework based on contour detection, complete the collection and training of basic shape samples, realize basic recognition in a simple non-interfering background;・Build a multi-condition testing environment, test the identification function and record results, errors, and failure scenarios, providing data support for algorithm optimization.

3.2 Electrical Connection & Hardware Preparation

・Complete the wiring deployment of all motors / servo motors, without any loose or incorrect connections, confirm and fix the positions of control and power supply pins, and standardize pin definitions;・Prepare core hardware such as motor / servo motor drivers, main controllers, power supply modules, etc., conduct on-off tests and pre-debugging of the electrical system, verify power conduction, circuit reliability, and basic hardware linkage.

3.3 Mechanical Structure Assembly & Motor Test

・Complete the production of optimized 3D printed structural components, successfully complete the assembly of motors / servo motors and structural components;・Conduct basic motor drive tests, discover that the motor cannot be driven normally, immediately start the fault troubleshooting work, and preliminarily complete the basic detection of hardware circuits, power supply modules.

3.4 Research on External Virtual Control Website

・Complete the preliminary framework setup of the external virtual control website, determine the wireless communication protocol with the main controller, achieve stable communication connection between the two;・Develop the basic control interface of the website, realize remote adjustment of single servo motor angles and simple combined action control, complete the stability test of signal transmission in the laboratory environment.

4. Observations & Findings

4.1 Core Findings of the Visual Recognition Module (Camera)

・The color recognition function is overall effective. In environments with sufficient light and a single background, the accuracy of recognizing conventional colors such as red, yellow, blue, and green is relatively high, and the color threshold division is reasonable;・The shape recognition is the core weakness. When there is slight occlusion, blurred contours, or background interference, it is very prone to misjudgment. It can only achieve basic recognition in a very simple background;・The recognition of black objects on a black desktop is highly likely to fail. The low contrast between the two makes the target contour unable to stand out, and the algorithm cannot accurately extract features, resulting in missed recognition and misrecognition;・Changes in light have a significant impact on the recognition effect. In low-light environments, the accuracy of color recognition decreases, the difficulty of extracting the contour of shapes increases, and the light adaptation ability of the camera and the algorithm is insufficient;・The basic image preprocessing algorithm cannot effectively filter noise and enhance contours, which indirectly affects the accuracy of shape recognition.

4.2 Findings of the Mechanical and Electrical Modules

・After optimization, the tolerance design of the 3D printed structural components is reasonable, and they can be precisely adapted to the motor / actuator. The mechanical assembly work was successfully completed;・The motor drive test found that the motor could not operate normally. The initial investigation did not find any issues such as loose wiring or power supply interruption. The cause of the fault has not been located yet;・The wiring and pin definitions of the motor / actuator are accurate, and the electrical system is conducting normally. The pre-verification response of each hardware module was normal, and the basic wiring faults were excluded.

4.3 Findings of the External Virtual Control Website

・The virtual website can realize the basic remote control of the servo motor. In the laboratory environment, the signal transmission is stable and there is no obvious delay. The communication link is reliable;・The operation interface of the website is not convenient. It only supports simple single-action control, and the control logic for complex combined movements has not been developed.

4. Challenges & Roadblocks

- 5.1 Core Challenges of the Visual Recognition Module (Camera)
- ・The shape recognition algorithm lacks robustness. The simple contour detection logic cannot handle actual scenarios such as background interference and slight occlusion, and lacks the ability to adapt to complex situations;・There is no effective recognition method for scenes with similar colors and low contrast. The existing algorithm relies on the differentiation of color and contour, but cannot solve the problem of identifying black objects on a black desktop;・The camera's environmental adaptability is poor. Manual parameter adjustment has limited effect. The algorithm lacks a light adaptation compensation mechanism, and the recognition effect significantly decreases in non-ideal lighting conditions;The image preprocessing process is

not complete. The basic algorithm cannot optimize the picture quality, and noise and blurriness problems directly affect the accuracy of shape recognition.

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- 5.2 Core Challenges of the Mechanical and Electrical Module
- ・After the motor assembly is completed, it cannot drive normally. The cause of the failure has not been located yet. A deep investigation needs to be conducted from multiple aspects such as the driver, control signals, and the motor itself;The motor drive failure has prevented the subsequent motion control tests and system integration work from progressing, becoming the main obstacle in the current project advancement.
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- 5.3 Other Module Challenges
- ・The external virtual control website lacks the combined motion control logic for the external devices. Function development needs to combine the motor drive and mechanical motion parameters. Further progress is temporarily impossible;The system integration of the visual recognition and motion control modules cannot be carried out. The motor drive failure has prevented the recognition signal from the camera from being transmitted to the actuating component, and the overall system integration work has been hindered.

6. Next Steps & Recommendations

・Optimize the shape recognition algorithm by introducing contour feature point extraction and matching technology, and adding core feature recognition such as corner points and edge length ratios, to enhance the recognition ability in complex scenarios and reduce the error rate

7. Conclusion

This week, the core of the biped robot project completed the setup and debugging of the visual recognition module (camera), achieving basic color recognition. However, the accuracy of shape recognition is insufficient, and the problem of frequent errors when recognizing black objects on a black surface still lies at the core of the optimization direction. After optimization, the 3D printing of structural components and their assembly were completed, but it was discovered that the motor could not be driven normally. Currently, fault investigation is underway and is expected to be resolved next week. At the same time, the wiring of the motor / steering gear, pin definitions, and pre-testing of the electrical system were successfully completed. An external virtual control website was implemented to achieve basic remote control of the steering gear. The motor drive failure has become the main obstacle in the progress of the project at present, preventing the conduct of motion control tests, system integration, and other related work. The overall progress of the project is controllable. Next week, the focus will be on optimizing the camera recognition algorithm, troubleshooting and repairing the motor drive failure, and immediately advancing the motor motion debugging, iterative improvement of the virtual website functions, and the linkage debugging of the visual recognition and motion control modules, gradually resolving the core technical difficulties at present and ensuring the project progresses as planned.